🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 3 · STANDARD 6.AT.3A

Determine Equivalent Ratios Using Graphs

Lesson 3-4 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

My Notes

Level 3 Enrichment

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can graph the values from a ratio table as points on the coordinate plane.

Language Objective: I can explain my graph using the words coordinate plane, ordered pair, origin, and proportional.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Graph Ratio Tables | Coordinate plane | Ordered pair | Linear pattern | Proportional | Origin | Ratio table

👆 *Tap any word to see what it means and a picture.*

Level 3: try the blanks from memory first, then check the bank.

1. Each row of a ratio table becomes one __________ __________ on the coordinate plane.

2. The flat grid formed by a horizontal and a vertical number line is the __________.

3. A set of two numbers (x, y) that names a point is an __________.

4. When graphed points form a straight line, they show a __________.

5. Two quantities that always keep the same ratio are ____________________.

6. The point (0, 0) where the two axes meet is the ____________________.

7. A table that lists equivalent ratios in rows or columns is a ____________________.

Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

1. Understand the Question

EXPLAIN

A food truck tracks how many tacos it sells and how much salsa it uses. After recording data for a week, the owner plots the points on a graph and notices they form a straight line. This helps the owner predict exactly how much salsa to prepare for any number of tacos?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

- [ ] **Graph Ratio Tables** — Plot the paired values from a ratio table as ordered pairs.
  Representar tablas de razones en una gráfica — Ubicar los valores emparejados de una tabla de razones como pares ordenados.
- [ ] **Coordinate plane** — A grid with a line going across and a line going up to plot points.
  Plano cartesiano — Una cuadrícula con una línea horizontal y una vertical para marcar puntos.
- [ ] **Ordered pair** — Two numbers (x, y) that tell where a point is on a grid.
  Par ordenado — Dos números (x, y) que dicen dónde está un punto en una cuadrícula.
- [ ] **Linear pattern** — Points that make a straight line on a grid.
  Patrón lineal — Puntos que forman una línea recta en una cuadrícula.
- [ ] **Proportional** — Two amounts that grow together at the same rate.
  Proporcional — Dos cantidades que crecen juntas al mismo ritmo.

Say it first: Say your idea to a partner or quietly to yourself before you write.
Di tu idea a un compañero o en voz baja antes de escribir.

Every point on the line has the same ratio — ____ taco to ____ ounces of salsa — so 10 tacos would use ____ ounces.
Di tu respuesta primero: Mi respuesta es ____ porque ____.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: I predict the points will line up because sundaes and sauce grow together.

Evidence: A strong answer chooses sundaes (x) and ounces of sauce (y), and predicts a straight line because each sundae always needs 2 more ounces.

Reasoning: This evidence connects the definition of graph ratio tables to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **Every point on the line has the same ratio — ____ taco to ____ ounces of salsa — so 10 tacos would use ____ ounces.**
- **My answer is ____.**
  Mi respuesta es ____.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
  Mi afirmación es ____.
- **I know because ____.**
  Lo sé porque ____.
- **So ____.**
  Entonces ____.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
  Mi afirmación es ____.
- **My math evidence is ____.**
  Mi evidencia matemática es ____.
- **This proves ____ because ____.**
  Esto demuestra ____ porque ____.

4. Check Your Explanation

- [ ] I answered the question.
  Respondí la pregunta.
- [ ] I used math evidence: numbers, an equation, a model, or a comparison.
  Usé evidencia matemática: números, una ecuación, un modelo o una comparación.
- [ ] I explained how the evidence proves my answer.
  Expliqué cómo la evidencia demuestra mi respuesta.
- [ ] I used at least two math words.
  Usé por lo menos dos palabras matemáticas.
- [ ] I reread complete sentences and punctuation.
  Volví a leer mis oraciones completas y la puntuación.

Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

Answer Key & Teacher Guide

1. **Notes 1:** Graph Ratio Tables
2. **Notes 2:** Coordinate plane
3. **Notes 3:** Ordered pair
4. **Notes 4:** Linear pattern
5. **Notes 5:** Proportional
6. **Notes 6:** Origin
7. **Notes 7:** Ratio table
8. **Watch for this mistake:** *A common mistake in Graph Ratio Tables is writing the ordered pair in the wrong order — putting the second table value first instead of matching (x, y) to the table's column order. For example, if a recipe table shows 2 cups of sugar for every 4 cups of flour, a student might plot (4, 2) instead of (2, 4), landing the point in a completely different spot on the grid. Before you plot, check: which quantity goes on the x-axis, and does your ordered pair start with that number?*
9. **Try It 1:** A. (3, 6) — *Cups (x) come first, scoops (y) second. 3 cups need 3 × 2 = 6 scoops, so the point is (3, 6).*
10. **Try It 2:** A. y = 2x — *Each cup needs 2 scoops, so y (scoops) is always 2 times x (cups): y = 2x. For 4 cups, y = 2 × 4 = 8.*